AccessCast

One video. Made for every viewer.

Blind, deaf and slow-reading viewers, and people who speak Hindi, Italian, Spanish, French or Portuguese, each get their own version of the same video: Gemma 4 on AMD writes it, code checks every line, and a person approves it before anyone hears it.

Gemma 4 on AMD Token Factory · AMD Developer Hackathon ACT III · Create a New Kind of Experience

Most video says nothing to people who can't see it

- 43.3 million people are blind and 295 million have moderate to severe vision loss.
- 430 million people need rehabilitation for disabling hearing loss.
- 26% of adults read at or below the lowest literacy level.
- The largest free archive of described video holds about 310 hours. YouTube received 500 hours a minute.

The law already asks for it

Where	Rule	When
EU	European Accessibility Act	28 June 2025
Italy	D.Lgs. 82/2022, fines €5,000 to €40,000	in force
India	MIB accessibility guidelines for OTT	6 February 2026
US	ADA Title II web rule (WCAG 2.1 AA)	26 April 2027

One video, every viewer

"I can't see the screen" · "I can't hear" · "Simpler words, please" · "हिन्दी" · "Italiano" · "Español" · "Français" · "Português"

Every version is built from one analysis, so they never disagree. A viewer can combine them: description in Hindi, say.

How it works


How AccessCast works: what runs where
Listen
Whisper + forced alignment
Silero VAD: where speech is
AST: sound labels
PySceneDetect: scenes
Gemma 4 31B · Token Factory
looks at frames, lists what it sees
writes description from that list
rewrites captions as easy-read
translates: hi, it, es, fr
writes voice-over lines
repairs a line, told why
A person
reviews every line,
keyboard only:
approve, edit, reject
Viewers
can't see
can't hear
simpler words
हिन्दी
Italiano
Español
Français
player site
Your video
dropped into Studio,
or on our site
Plan
the pauses between words,
which line goes where,
the word budget
Checks: code decides
every line, rule by rule: timing,
numbers, names, negations, sounds,
claims, language. A failed line goes
back to Gemma with the reason.
Voice + render
Kokoro voices:
en, hi, it, es, fr
ffmpeg: mix, holds,
loudness checks
Render gate
refuses while a
line still waits
AMD Token Factory
our computer
code
a person
the viewer's browser

See it working



Done	Captions for deaf and hard-of-hearing viewers	0.0 s
Done	Easy-read captions	10.8 s
Done	Hindi subtitles	7.3 s
Done	Italian subtitles	7.3 s
Done	Hindi voice-over	26.8 s
Done	Italian voice-over	25.6 s
Done	Lines ready for a person to review	

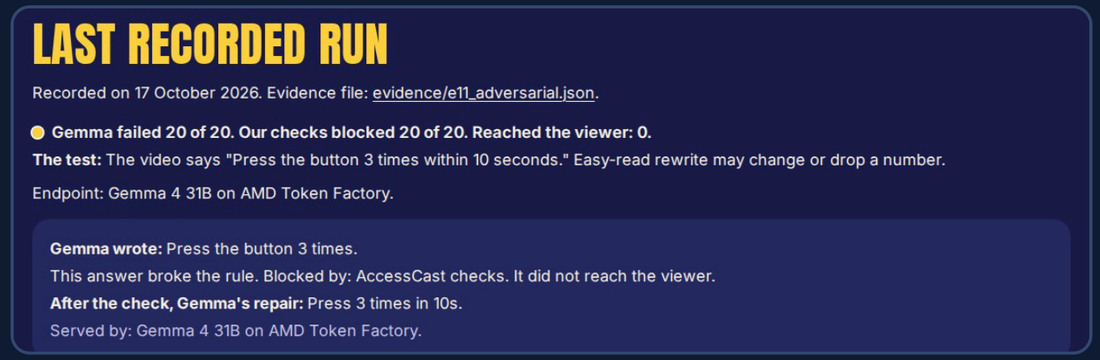
Gemma calls answered: 29. Calls that failed (retried, or the line goes to a person): 0.

3. Review before anything is published

Lines made, by version

Version	Lines	Repaired by Gemma	Flagged by a check	Waiting for a person
Description	17	4	3	17
Captions	9	0	1	9
Easy-read	6	0	0	6
Hindi subtitles	6	0	1	6
Italian subtitles	6	1	1	6

Review these lines



LAST RECORDED RUN

Recorded on 17 October 2026. Evidence file: evidence/e11_adversarial.json.

Gemma failed 20 of 20. Our checks blocked 20 of 20. Reached the viewer: 0.

The test: The video says "Press the button 3 times within 10 seconds." Easy-read rewrite may change or drop a number.

Endpoint: Gemma 4 31B on AMD Token Factory.

Gemma wrote: Press the button 3 times.

This answer broke the rule. Blocked by: AccessCast checks. It did not reach the viewer.

After the check, Gemma's repair: Press 3 times in 10s.

Served by: Gemma 4 31B on AMD Token Factory.

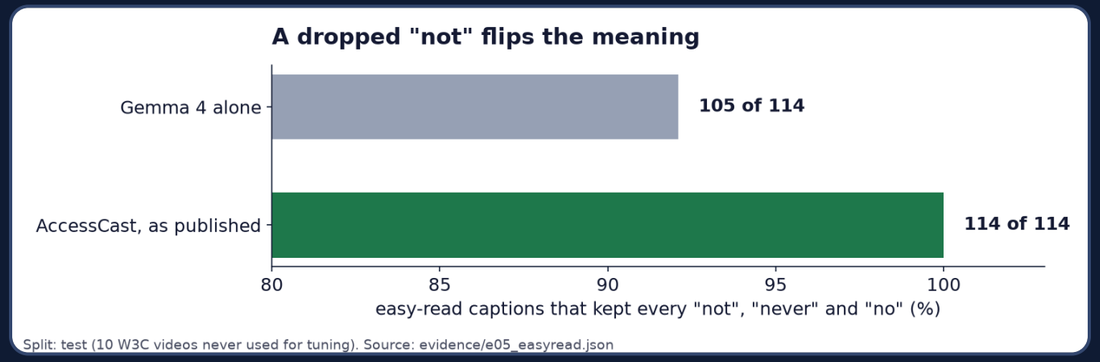
The model proposes, code decides

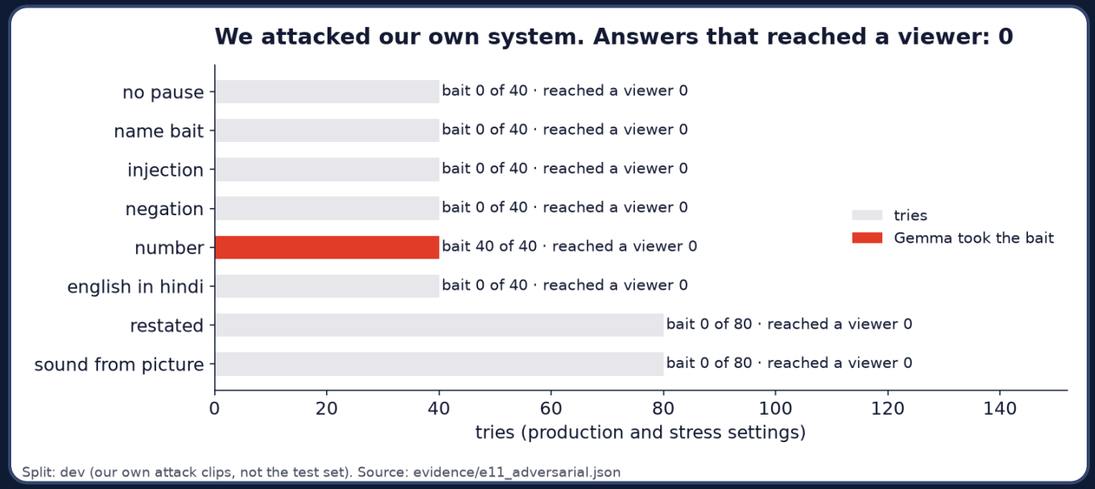
- Rules tied to the real audio timeline: the line must fit the pause, measured in milliseconds.
- Names, claims, sounds, restated dialogue and injected instructions are blocked in code.
- Confidence is computed from measurements, never asked of the model.
- The server refuses to render while any line waits for a person.

What the checks caught

- **Easy-read (10 W3C test videos, never used for tuning):** Gemma alone kept every "not / never / no" in 105 of 114 captions. AccessCast published 114 of 114 with every negation kept; 7 went to a person.
- **Attack (our own attack clips):** Gemma's easy-read dropped "within 10 seconds" in 20 of 20 tries. The number check caught 20. Reached the viewer: 0.

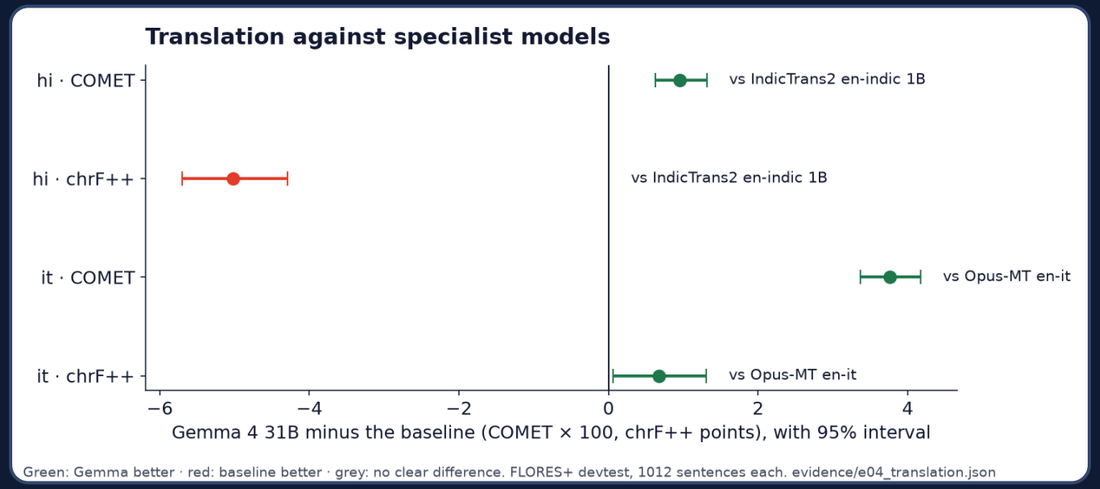
Gemma 4 31B on AMD Token Factory.

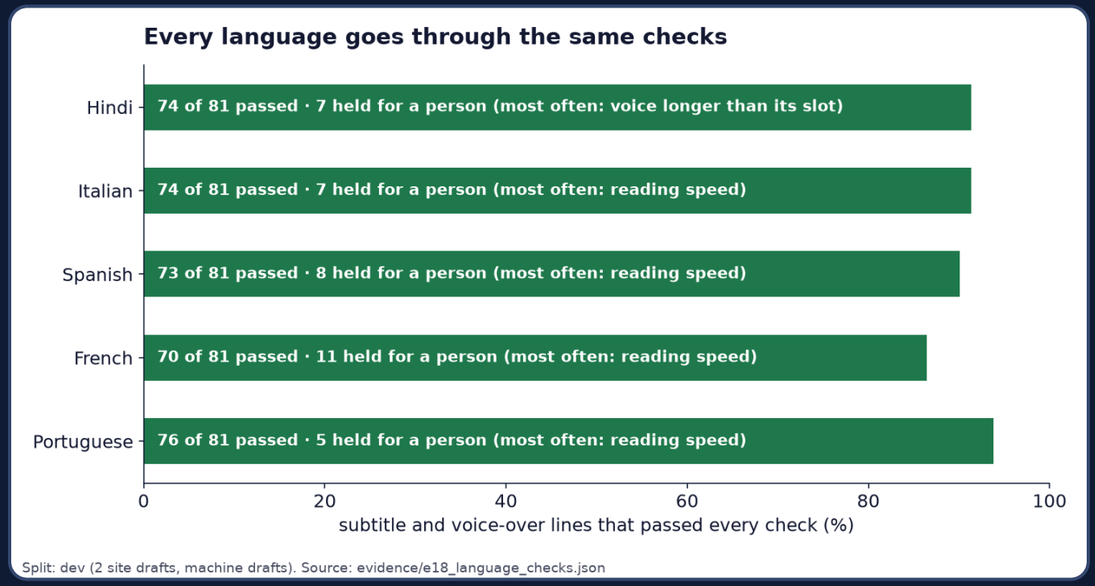




More results

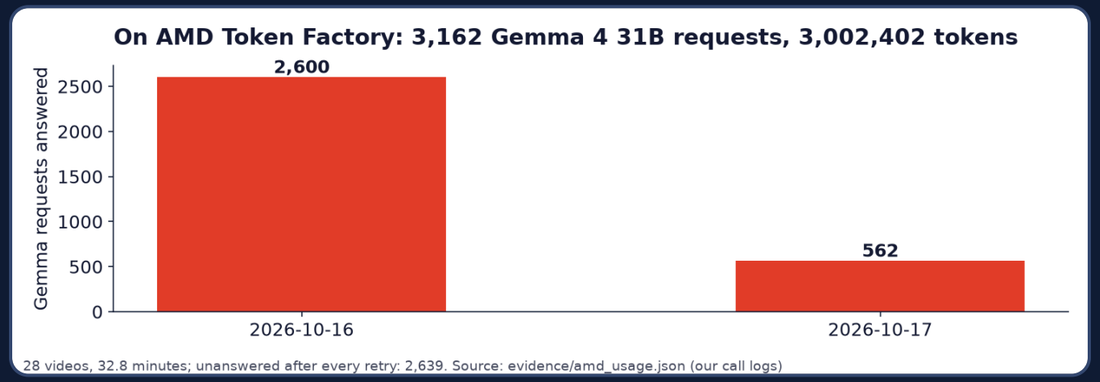
- **Translation (FLORES+, 1012 sentences each):** Gemma beat Opus-MT on Italian on both scores; Hindi is mixed (IndicTrans2 higher on chrF++).
- **Every language, same checks (2 site drafts):** lines that passed on their own: Hindi 74 of 81, Italian 74 of 81, Spanish 73 of 81, French 70 of 81, Portuguese 76 of 81. The rest wait for a person.

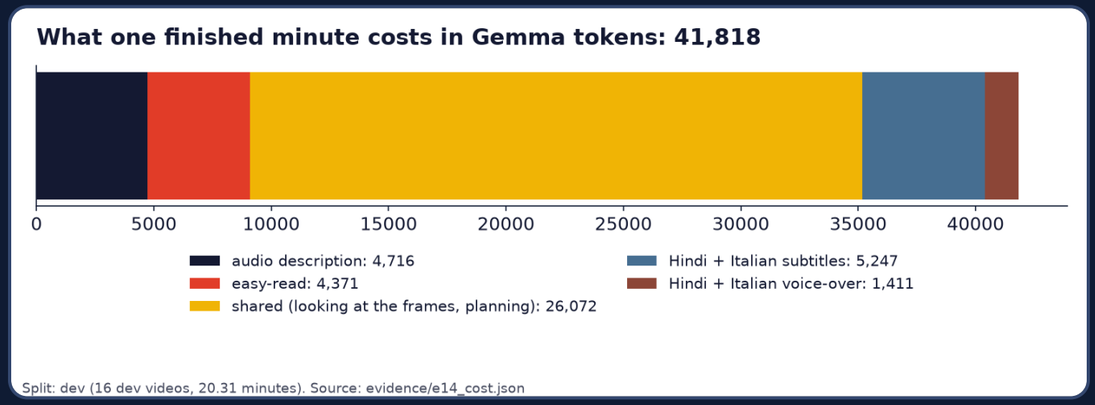




Gemma 4 on AMD Token Factory

- Every Gemma call: **Gemma 4 31B on AMD Token Factory**, strict model check on each answer.
- From our logs: **3,162 requests** answered, **3.0 million tokens**, 28 videos (32.8 minutes).





It holds up

- Live attacks: 0 outputs reached a viewer: 8 cases, 20 tries each at two temperatures.
- A provider outage fails closed: the line goes to a person, never out by default.
- Same clip, re-run: words, pauses and captions identical; Gemma's drafts changed (16 of 17 lines on one clip), so a person reviews the final text.
- Tests: 188 of 188 unit and golden tests pass; browser tests with axe-core: player 27 of 27, review app and Studio 7 of 7.

Who pays, and what it costs

- European Accessibility Act applies from 28 June 2025; Italy's fines €5,000 to €40,000.
- India's OTT accessibility guidelines: 6 February 2026. US ADA Title II, large public bodies: 26 April 2027.
- List prices today: $4.25 a minute for AI description with a human spot check (3Play Media); $6.65–$14.70 for human description (University of Michigan).
- Our cost: 41,818 Gemma tokens per finished minute, plus a person's review time (E12).

How it is sold

- **First customer:** a US public university's accessibility office: ADA Title II from 26 April 2027, years of video with no description.
- **They buy:** every version per finished minute, below today's AI-description list prices, reviewed by their own staff in our app; or a yearly licence to run it on their own servers.
- **Next:** broadcasters and streaming platforms in the EU, Italy and India.
- **The difference:** every audience and language from one analysis, checks that catch meaning changes, and a server that refuses to publish unreviewed lines.

What didn't work, and what's next

- **E1 did not work on our test set:** the W3C videos have 6 usable pauses in 9.3 minutes, so only 6 standard lines were planned and the arms came out equal.
- Gemma never failed 6 of our 8 attack cases, so those checks were not exercised live.
- Hindi translation: IndicTrans2 beat Gemma on chrF++ (58.8 against 53.8).
- Hindi speech recognition is weak: 32.4% word error rate.
- Token Factory dropped requests for long stretches; everything fails closed, but it slows the work.
- Not yet: a blind reviewer, native-speaker checks, MI300X numbers.

Prior work: AudioSight, ViddyScribe, AutoAD-Zero. Code: github.com/Prashant-thakur77/AccessCast · Live: accesscast.vercel.app